

Opportunity areas around video surveillance technology

Krishna foresees video surveillance-as-a-service emerging as a huge opportunity area in India.

He explains: "With the coming of 3G networks, huge bandwidths will soon be available to telecom operators and high-quality video streaming will become a reality. As a result, the operators will soon be looking at applications that enable the use of this available bandwidth. In this scenario, video surveillance as a service may find traction. Operators these days lease out routers and DSL modems to allow users to connect to the Internet. Along with these devices, they can even install a network of cameras for surveillance (which are available at reasonably low cost nowadays) and store the footage at their data centre. Users can have access to this recording post-event or view it in real time."

Optical sensors (or cameras) can also be replaced with other sensors such as IR or thermal using the same analytics for alert generation.

Optimised to save power. The cameras of the Silvan video surveillance system are equipped with an on-board battery that provides up to 30 days of power backup. The patented algorithms embedded in the cameras help them conserve power. For instance, in the event of power failure, when the camera is running on the battery, it may choose to switch off a few functions that are not required. Also, the algorithm may put the camera to sleep in the absence of any movement within the surveillance area and bring it up to function when there is movement.

Image capture even in the dark. The surveillance system can capture video images at night without the need for an expensive day/night camera. The cameras have an integrated passive infrared (PIR) sensor along with LED lighting and a standard CMOS sensor. The PIR sensor detects human movement, which triggers the operation of

the camera. The in-built LEDs glow to illuminate the scene and provide a clear and full-colour video output. In the event of a power cut too, the camera continues to capture images. Once the network restores, the stored data is streamed to the master camera and the collected feed is then streamed via the master camera to the visual medium.

What's most innovative

The company has filed four patents on innovations related to the video surveillance system. The patents cover operational analytics that enable optimal utilisation of power, storage and network bandwidth, and distributed video surveillance. Claims Krishna, "Algorithms that we have built for power optimisation and intelligence distribution across the network is something which has not been reported so far by any other company."

Tech challenges

Krishna recalls and shares a couple of challenges that the team faced during the process of product development.

"A team of twelve engineers (hardware and software) worked on this technology for over a year and most of the challenges related to hardware and software co-design. Close collaboration between the hardware and software teams was essential. This requirement was addressed by ensuring that all design decisions were taken jointly and all features were analysed and reviewed jointly to understand the impact of each new addition on hardware and software components."

Another challenge related to the required expertise and skill-set of the team of engineers. "Since the underlying operating system in use is Linux, the biggest challenge was to put in place a team that was well-trained and knowledgeable in embedded Linux. This continues to be a challenge till date," says Krishna.

Going forward...

Currently, each camera in the Silvan video surveillance system can do only one type of event detection, based on the algorithm that specifies the event for which the alert is to be generated. But going forward, the Silvan team is working on making the cameras smarter by embedding intelligence that enables each camera to handle complex scenarios and generate alerts for up to three events.

In the face of the security threats that surround us, this is surely going to be a solution to look forward to! ●

The author is an assistant editor at EYF

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